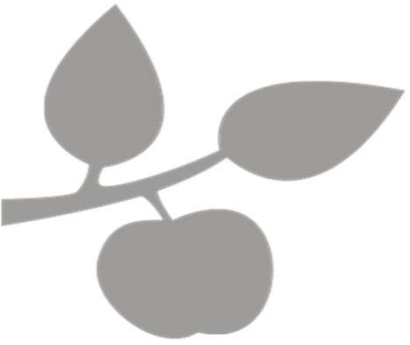




DM583

Data Mining

Overlapping Partitioning Clustering

Partition Matrix Revisited

- It is a matrix with k rows (no. of clusters) and N columns (no. of observations) in which entry μ_{ij} stands for the membership degree of the j th observation ($\mathbf{x}_j$) to the i th cluster ($\mathbf{C}_i$)
- In case of **hard (non-overlapping)** partitioning algorithms, each observation must belong to exactly one cluster, i.e:

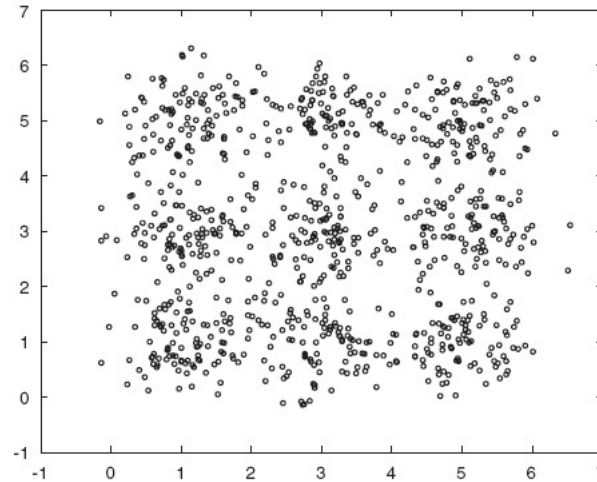
$$\mu_{ij} \in \{0,1\}$$

$$\sum_i \mu_{ij} = 1 \quad \forall j$$

$$\mathbf{U}(\mathbf{X}) = \begin{matrix} & \text{observations} \\ \begin{matrix} \mu_{11} & \mu_{12} & \cdots & \mu_{1N} \\ \mu_{21} & \mu_{22} & \cdots & \mu_{2N} \\ \vdots & & \ddots & \vdots \\ \mu_{k1} & \mu_{k2} & \cdots & \mu_{kN} \end{matrix} & \left. \vphantom{\begin{matrix} \mu_{11} \\ \mu_{21} \\ \vdots \\ \mu_{k1} \end{matrix}} \right\} \text{clusters} \end{matrix}$$

Hard vs Overlapping Partitioning Clustering

- Partitioning algorithms such as k-means, k-medoids and others produce a **non-overlapping partition** of the data:
 - Each object fully belongs to exactly one cluster
 - Usually, we refer to this type of partition as **Hard** or **Crisp**
- However, many problems involve not clearly delineated clusters that cannot be properly represented in this way
- In other words, there are situations in which the data comprise categories that overlap each other at different levels
- For instance:



Overlapping Partitioning Clustering Algorithms

- **Overlapping clustering algorithms** exist that can cope with these situations
- They can be subdivided into three categories, depending on the type of partition that they produce:
 - **Soft:** Objects can (fully) belong to more than one cluster
 - **Fuzzy:** Objects can partially belong to multiple clusters
 - Cluster membership is a matter of degree (possibly null)
 - **Probabilistic:** There is uncertainty of a probabilistic nature on the association between objects and clusters
 - Cluster membership is interpreted as likelihood

Fuzzy and Probabilistic Partitions

- **Fuzzy Partition Matrix:** real-valued membership values, $\mu_{ij} \in [0,1]$

$$\mathbf{U}(\mathbf{X}) = \begin{bmatrix} \mu_{11} & \mu_{12} & \cdots & \mu_{1N} \\ \mu_{21} & \mu_{22} & \cdots & \mu_{2N} \\ \vdots & & \ddots & \vdots \\ \mu_{k1} & \mu_{k2} & \cdots & \mu_{kN} \end{bmatrix}$$

- **Probabilistic Partition Matrix:** $\mu_{ij} \in [0,1]$ interpreted as probabilities
 - Therefore: $\sum_i (\mu_{ij}) = 1 \quad \forall j$

Fuzzy and Probabilistic Partitions

- **Example (Fuzzy):**

$$\mathbf{U}(\mathbf{X}) = \begin{bmatrix} 1 & 0.1 & 0.5 & 0.1 \\ 0 & 0.1 & 0.5 & 0.9 \end{bmatrix}$$

- **Example (Fuzzy / Probabilistic):**

$$\mathbf{U}(\mathbf{X}) = \begin{bmatrix} 1 & 0.7 & 0.5 & 0.1 \\ 0 & 0.3 & 0.5 & 0.9 \end{bmatrix}$$

Model-Based Clustering

- In **Model-Based Clustering** we assume that a dataset is a sample from a population, then we make formal statistical assumptions about this population and use the sample to fit a *parametric model*
- The most well-known and commonly used model is the **Gaussian Mixture Model (GMM)**, which assumes the data is drawn from a (multi-normal) *mixture distribution*
 - This is a probabilistic model for overlapping partitioning clustering

Model-Based Clustering

- GMM models can be fit using the EM Algorithm
- **EM (Expectation Maximization)** is a probabilistic modelling procedure based on the principle of **Maximum Likelihood Estimation (MLE)**

GMMs

- Gaussian Mixture Models are described by the following probability density function p :

$$p(\mathbf{x}_j) = \sum_{i=1}^k \pi_i \mathcal{N}(\mathbf{x}_j | \mathbf{v}_i, \Sigma_i)$$

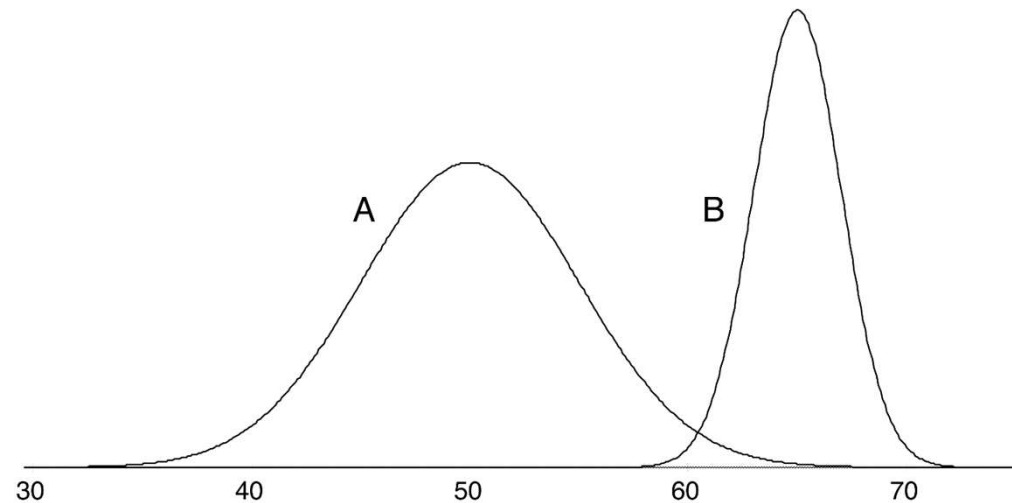
- $\mathbf{x}_j$ is a vector of n real-valued random variables (a data object in $\mathbb{R}^n$)
- $\mathcal{N}$ is a multivariate Gaussian (same dimension as the objects)
 - $\mathbf{v}_i$ is the centre of the i th Gaussian (vector of same dimension as $\mathbf{x}_j$)
 - Σ_i is the covariance matrix of the i th Gaussian
- π_i is the prior probability associated with the i th Gaussian
- k is the number of Gaussians

GMMs: 1-dimensional Example

Objects:

A	51	B	62	B	64	A	48	A	39	A	51
A	43	A	47	A	51	B	64	B	62	A	48
B	62	A	52	A	52	A	51	B	64	B	64
B	64	B	64	B	62	B	63	A	52	A	42
A	45	A	51	A	49	A	43	B	63	A	48
A	42	B	65	A	48	B	65	B	64	A	41
A	46	A	48	B	62	B	66	A	48		
A	45	A	49	A	43	B	65	B	64		
A	45	A	46	A	40	A	46	A	48		

Model:



$$\nu_A=50, \sigma_A=5, \pi_A=0.6 \quad \nu_B=65, \sigma_B=2, \pi_B=0.4$$

EM for GMMs

- The following quantity plays a fundamental role when fitting GMMs with EM:

$$\mu_{ij} = \frac{\pi_i \mathcal{N}(\mathbf{x}_j | \mathbf{v}_i, \Sigma_i)}{\sum_{l=1}^k \pi_l \mathcal{N}(\mathbf{x}_j | \mathbf{v}_l, \Sigma_l)}$$

- μ_{ij} is the **posterior probability** that an observed value $\mathbf{x}_j$ was generated by the i th cluster (i.e., by the i th multivariate Gaussian component)
 - The probability associated with the i th component *conditioned* to $\mathbf{x}_j$
 - It trivially follows from the Bayes theorem

EM for GMMs

- Another fundamental quantity (**likelihood**):
 - Given a sample (dataset) $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ with N i.i.d. observations $\mathbf{x}_i \in \mathbb{R}^n$, their joint distribution is:

$$\blacksquare \quad p(\mathbf{X}) = p(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N) = \prod_{j=1}^N p(\mathbf{x}_j) = \prod_{j=1}^N \sum_{l=1}^k \pi_l \mathcal{N}(\mathbf{x}_j | \mathbf{v}_l, \Sigma_l)$$

- This is the likelihood that a sample $\mathbf{X}$ will be observed/drawn from the Gaussian mixture distribution in question, with parameters:
 - $\Sigma = \{\Sigma_1, \dots, \Sigma_k\}$, $\mathbf{v} = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ and $\pi = \{\pi_1, \dots, \pi_k\}$
 - For this reason, we also use the notation $p(\mathbf{X} | \pi, \Sigma, \mathbf{v})$

EM for GMMs

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 - For this reason, we also use the notation $p(\mathbf{X} | \pi, \Sigma, \mathbf{v})$
 - For mathematical convenience, a log transformation is applied to replace products with sums, resulting in the **log-likelihood** function:

$$\ln(p(\mathbf{X} | \pi, \Sigma, \mathbf{v})) = \sum_{j=1}^N \ln \left(\sum_{l=1}^k \pi_l \mathcal{N}(\mathbf{x}_j | \mathbf{v}_l, \Sigma_l) \right)$$

EM for GMMs

- Maximising the likelihood (or, equivalently, the log-likelihood) can be seen as maximising the agreement/match between the sample and the model
- EM (Dempster et al., 1977) is an optimisation algorithm that aims to maximise the (log) likelihood function in 2 steps:
 - **Step E** (Expectation)
 - Evaluate the posterior probabilities μ_{ij} ($i = 1, \dots, k; j = 1, \dots, N$)
 - from the N observations $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ and the current model, given by parameters $\Sigma = \{\Sigma_1, \dots, \Sigma_k\}$, $\mathbf{v} = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ and $\pi = \{\pi_1, \dots, \pi_k\}$
 - **Step M** (Maximisation)
 - Adjust the model parameters to locally maximise the log-likelihood function

EM for GMMs

- **Step E** (Expectation):

- Evaluate the posterior probabilities μ_{ij} ($i = 1, \dots, k; j = 1, \dots, N$)

$$\mu_{ij} = \frac{\pi_i \mathcal{N}(\mathbf{x}_j | \mathbf{v}_i, \mathbf{\Sigma}_i)}{\sum_{l=1}^k \pi_l \mathcal{N}(\mathbf{x}_j | \mathbf{v}_l, \mathbf{\Sigma}_l)}$$

$$\mathcal{N}(\mathbf{x}_j | \mathbf{v}_i, \mathbf{\Sigma}_i) = \frac{1}{(2\pi)^{n/2} \det(\mathbf{\Sigma}_i)^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}_j - \mathbf{v}_i)^T \mathbf{\Sigma}_i^{-1}(\mathbf{x}_j - \mathbf{v}_i)\right\}$$

EM for GMMs

- **Step E** (Expectation):

- Evaluate the posterior probabilities μ_{ij} ($i = 1, \dots, k; j = 1, \dots, N$)

$$\mu_{ij} = \frac{\pi_i \mathcal{N}(\mathbf{x}_j | \mathbf{v}_i, \mathbf{\Sigma}_i)}{\sum_{l=1}^k \pi_l \mathcal{N}(\mathbf{x}_j | \mathbf{v}_l, \mathbf{\Sigma}_l)}$$

$$\mathcal{N}(\mathbf{x}_j | \mathbf{v}_i, \mathbf{\Sigma}_i) = \frac{1}{(2\pi)^{n/2} \det(\mathbf{\Sigma}_i)^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x}_j - \mathbf{v}_i)^T \mathbf{\Sigma}_i^{-1} (\mathbf{x}_j - \mathbf{v}_i) \right\}$$

Mahalanobis distance (squared)
from observation to cluster center

EM for GMMs

- **Step M** (Maximisation):

- Adjust the model parameters

$$\left\{ \begin{array}{ll} \mathbf{v}_i = \frac{1}{N_i} \sum_{j=1}^N \mu_{ij} \mathbf{x}_j & \Rightarrow \text{weighted centroid} \\ \boldsymbol{\Sigma}_i = \frac{1}{N_i} \sum_{j=1}^N \mu_{ij} (\mathbf{x}_j - \mathbf{v}_i)(\mathbf{x}_j - \mathbf{v}_i)^T & \Rightarrow \text{weighted covariance (*)} \\ \pi_i = \frac{N_i}{N} & \Rightarrow \text{relative "responsibility" of the } i\text{th cluster (Gaussian)} \\ N_i = \sum_{j=1}^N \mu_{ij} & \Rightarrow \text{absolute "responsibility" of the } i\text{th cluster (Gaussian)} \end{array} \right.$$

EM for GMMs

- **Algorithm:**

- 1. **Initialisation** (e.g. via k-means)

- prototypes $\mathbf{v}_i$ = resulting k-means centroids
 - covariances Σ_i = sample covariance matrices of the resulting clusters
 - probabilities μ_{ij} (for N_i and π_i) = resulting (hard) partition matrix

- 2. **Step E**

- 3. **Step M**

- 4. **Evaluate Stopping Criterion**

- e.g. Log-likelihood function, no. iterations, ...

- 5. **Stop or Return to Step 2**

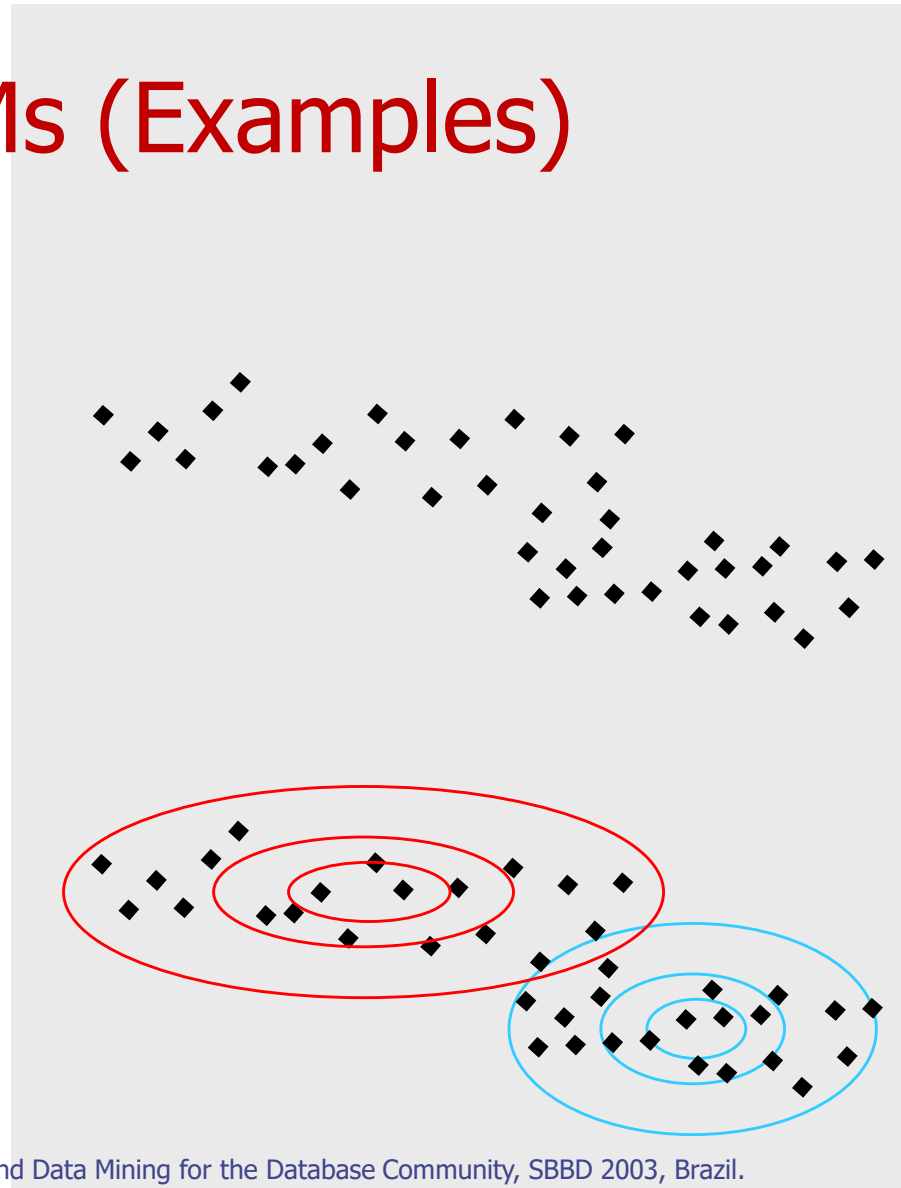
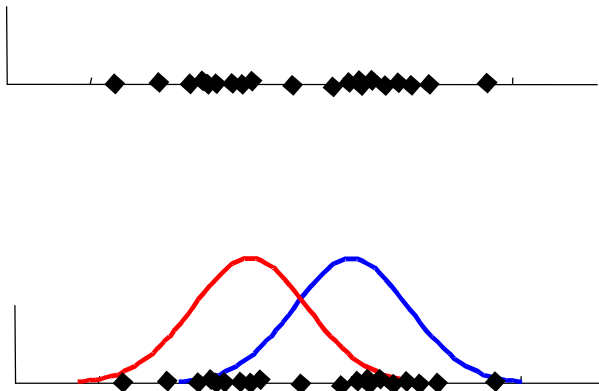
EM (GMMs) Summary

- Returns a **probabilistic model**
- Based on a **parametric assumption**
 - Although with a flexible / powerful parametric distribution
- Applicable to **real-valued feature vector data** only
- **Dimensionality sensitive**
 - Number of parameters to be fit grows quadratically with the number of dimensions n
 - Amount of data required to estimate covariance matrices grows at a fast pace with n
 - Prone to numerical issues and overfitting
 - Computing the inverse of the covariance matrices Σ_i requires $O(n^3)$ time
 - There are variants and simplified versions that are more robust and/or faster (e.g., see **MCLUST**)
- **Non-hierarchical (number of clusters required as input)**
- Subject to **local optima (local minima/maxima)**

EM (GMMs) × k-Means

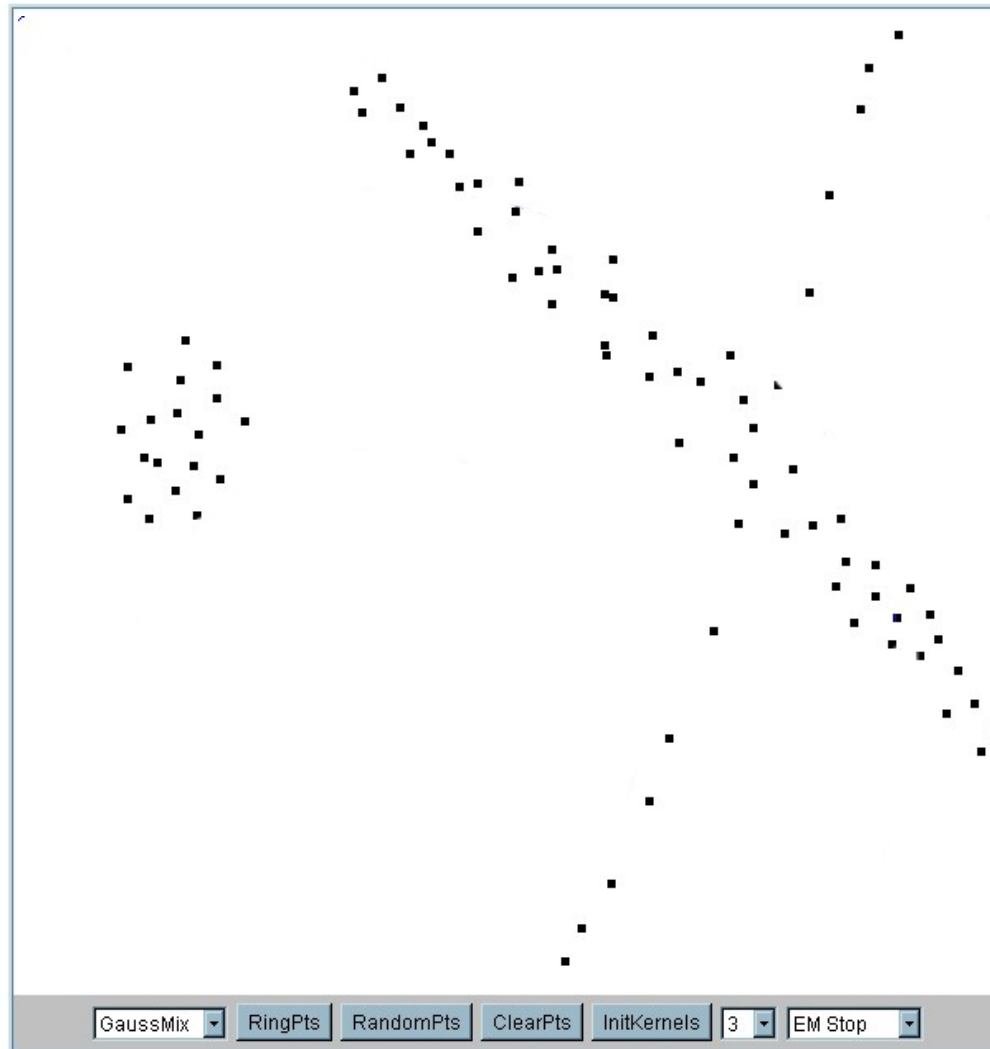
- EM provides much more information about the data
 - PDF model describing the distributions of clusters
 - Posterior probabilities for each observation
 - Outliers...
- It can model elongated, ellipsoidal clusters with arbitrary covariances
- However, this comes with a price:
 - GMMs have a much larger number of parameters to be fit (more data needed, etc.)
 - Computing the inverse of the covariance matrices Σ_i requires $O(n^3)$ time
 - There are variants and simplified versions that are more robust and/or faster (e.g., see **MCLUST**)
- k-means can be shown to be a particular limit of EM-GMM
 - Both are subject to local minima

EM-GMMs (Examples)



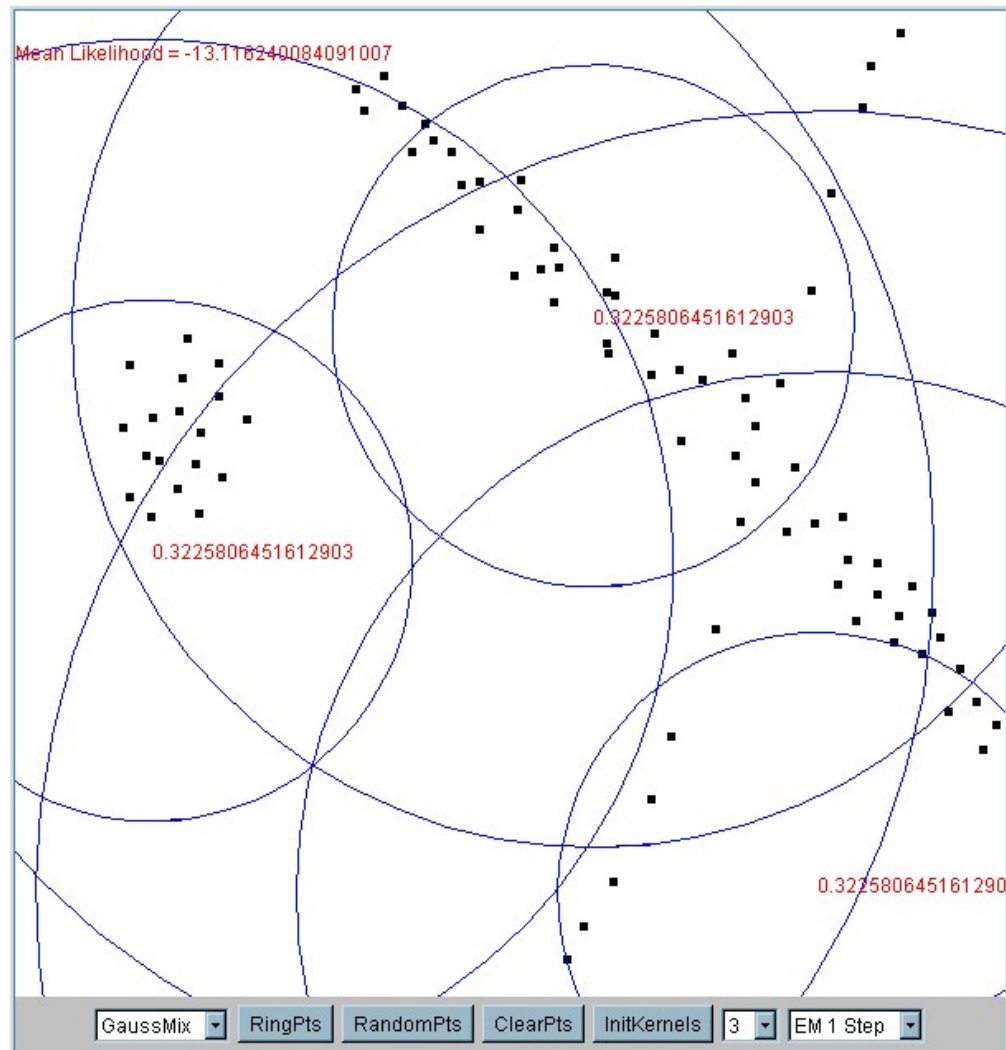
Keogh, E. A Gentle Introduction to Machine Learning and Data Mining for the Database Community, SBB D 2003, Brazil.

Example (step-by-step)



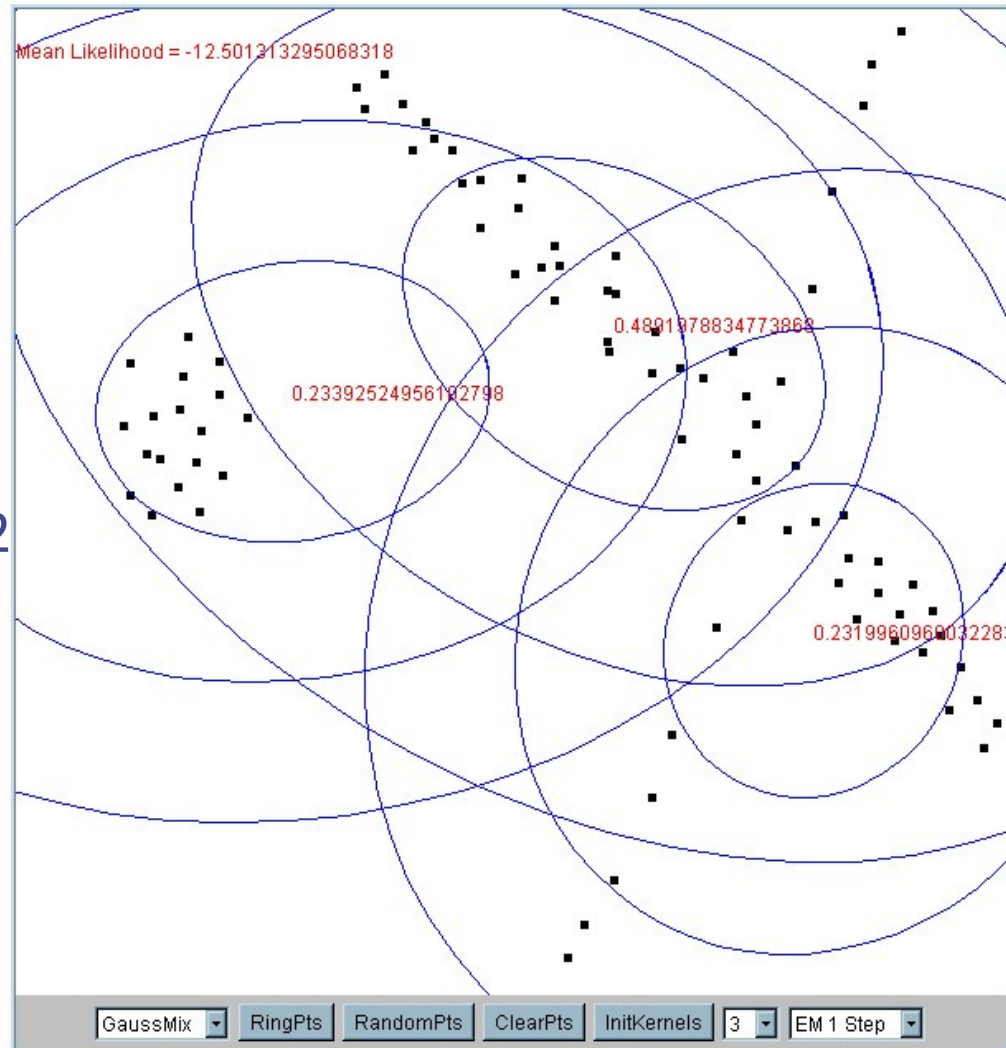
Keogh, E. A Gentle Introduction to Machine Learning and Data Mining for the Database Community, SBBD 2003, Brazil.

Iteration 1



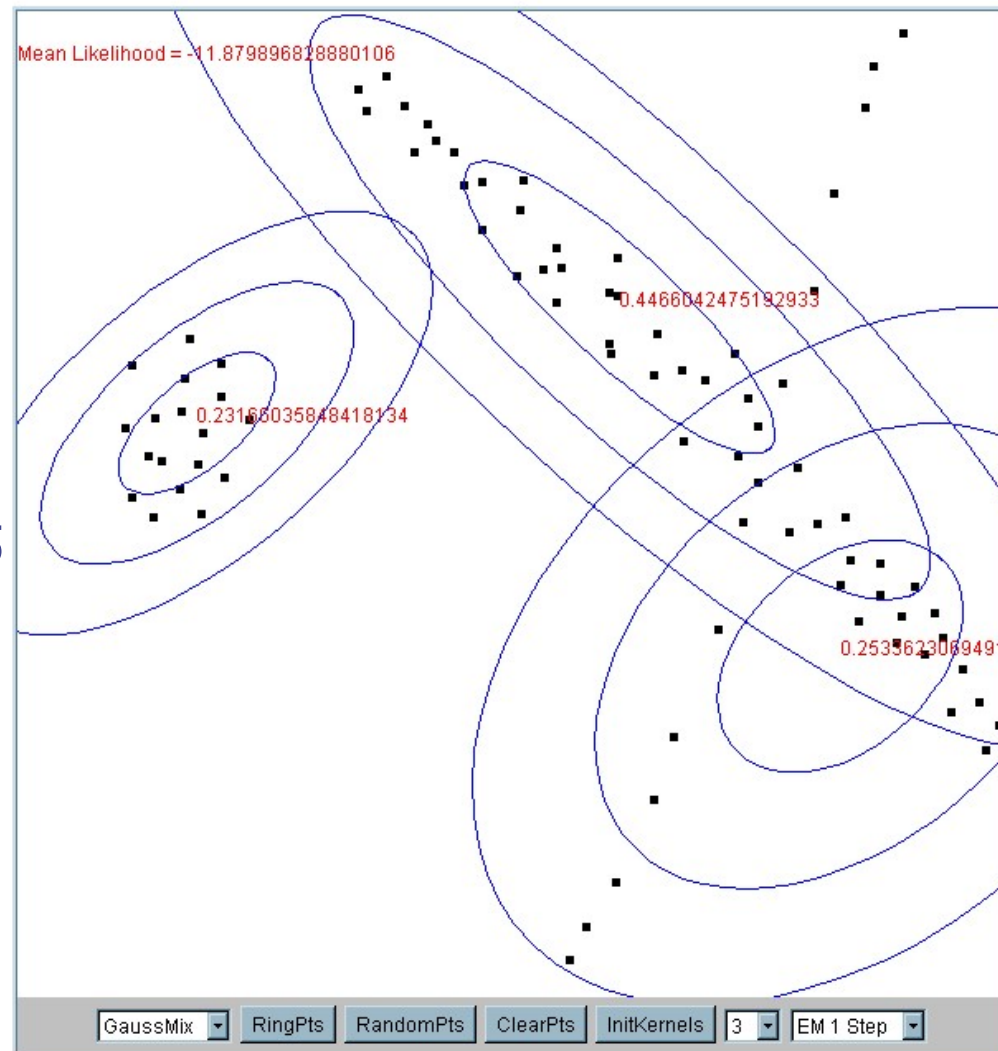
Keogh, E. A Gentle Introduction to Machine Learning and Data Mining for the Database Community, SBBD 2003, Brazil.

Iteration 2



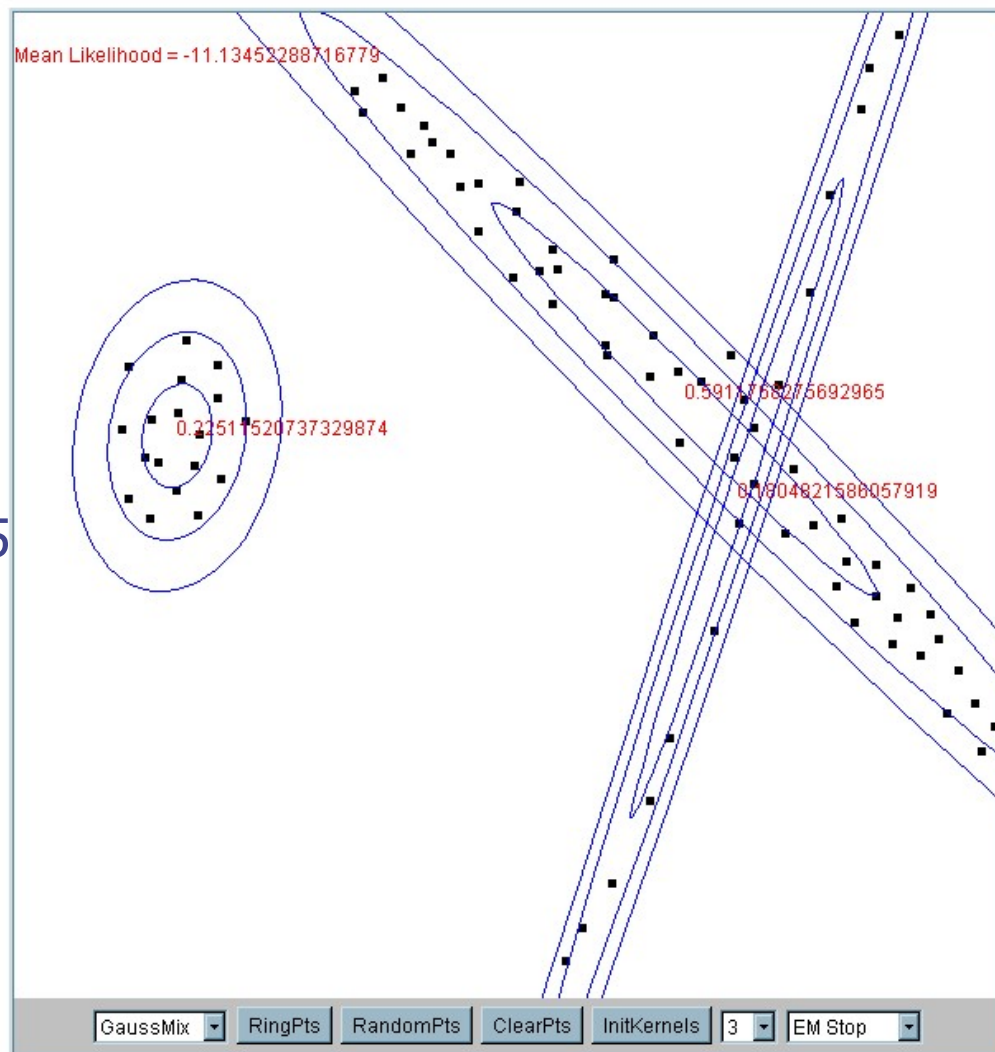
Keogh, E. A Gentle Introduction to Machine Learning and Data Mining for the Database Community, SBBD 2003, Brazil.

Iteration 5



Keogh, E. A Gentle Introduction to Machine Learning and Data Mining for the Database Community, SBBB 2003, Brazil.

Iteration 25



Keogh, E. A Gentle Introduction to Machine Learning and Data Mining for the Database Community, SBBD 2003, Brazil.

Exercise

Object ID	x_1
1	-1.31
2	-0.43
3	0.34
4	3.57
5	2.76
6	0.30
7	9.06
8	4.45
9	2.87
10	4.42

- Manually perform EM in this dataset ($n = 1$, $N = 10$), with $k = 2$. Start from arbitrary/random clusters
- Illustrate the results graphically

MCLUST Framework [OPTIONAL]

TABLE 1. Parametrizations of the covariance matrix Σ_k in the Gaussian model and their geometric interpretation. The models shown here are those discussed in Banfield and Raftery [2].

Σ_k	Distribution	Volume	Shape	Orientation	Reference
λI	Spherical	Equal	Equal	NA	1, 2, 5, 20
$\lambda_k I$	Spherical	Variable	Equal	NA	2, 5
$\lambda D A D$	Ellipsoidal	Equal	Equal	Equal	2, 5, 21, 22
$\lambda_k D_k A_k D_k$	Ellipsoidal	Variable	Variable	Variable	2, 5, 22
$\lambda D_k A D_k$	Ellipsoidal	Equal	Equal	Variable	1, 2, 5
$\lambda_k D_k A D_k$	Ellipsoidal	Variable	Equal	Variable	2, 5

- C. Fraley and Adrian E. Raftery "How many clusters? Which clustering method? Answers via model-based cluster analysis" *The computer journal* 41.8 (1998): 578-588
- C. Fraley and Adrian E. Raftery "MCLUST: Software for model-based cluster analysis" *Journal of classification* 16.2 (1999): 297-306.

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